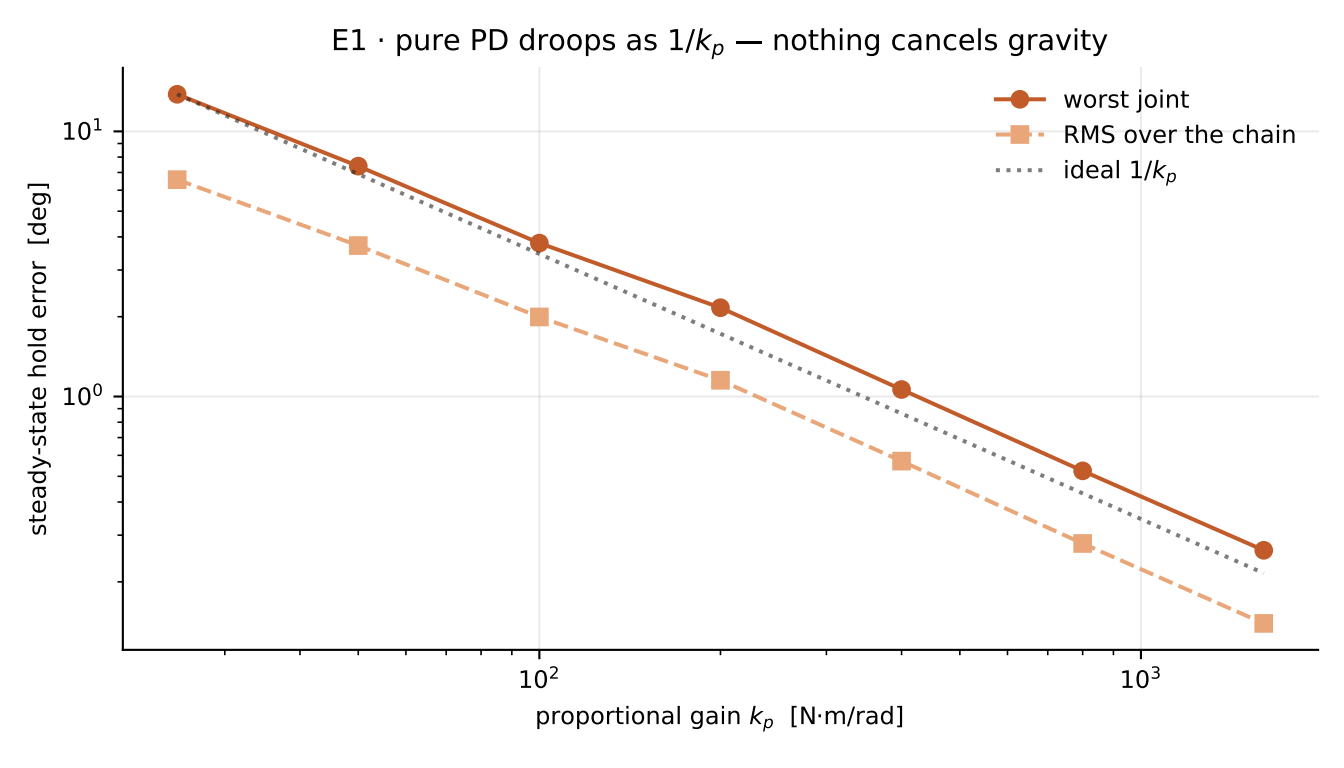
E1 · pure PD droops as 1/k_p — nothing cancels gravity
worst joint
RMS over the chain
ideal 1/k_p
steady-state hold error [deg]
10^1
10^0
10^2
10^3
proportional gain k_p [N·m/rad]